

Object: Instructions for filling and bleeding the hydraulic system "M"

The aim of these instructions is to remove, by the bleeding procedure, undissolved air inside the hydraulic system during the filling process. The **hydraulic system is to be filled** as follows:

1. Install the spring cylinder without making the connection to the end of the rod. In order to achieve the slanting position, the spring cylinder can be held by means of the stock absorber.
2. Connect the hoses of the hydraulic system to the spring cylinder.
3. Dismantle the miniature measuring connections and install the ball valves, including the hose connections, to the spring cylinders.
4. Fill the hydraulic cabinet HSLM with hydraulic oil class VG 68 with the help of an oil pump and a particulate filter.
5. Place clean oil drums under each spring cylinder and make a return connection for any excess oil overflowing from the operating pressure side.
6. Fill all hydraulic accumulators at a sufficiently high pressure, so that lifting of the bladder or the piston from the accumulator base during filling is prevented.
7. Open the ball valve (bleeding valve) on the spring cylinder furthest removed from the HSLM so that air can escape during filling. During this process the piston must be in the retracted position. The valves for the remaining cylinders must stay closed.
8. Using the on-site switch in the hydraulic cabinet, piping system "A" of the hydraulic system is filled with oil.
As soon as oil starts escaping, the pump must be switched off and the ball valve must be closed. Using the bleeding screws, the bladder accumulator belonging to the filled spring cylinders that are not directly connected to the circuit pipe are to be bled. Then the ball valve on the rod side has to be opened again and the pump has to be restarted.
Then, in order to be sure that no air has been left in the hydraulic pipes, approx. 4 times the quantity has to be pumped from the operating pressure chamber of the spring cylinder via the ball valve and the pipe into the oil drum. Once this process is finished, first the pump must be switched off, and then the ball cock on the cylinder must be closed.

9. The one after the other, the cylinders furthest removed from the HSLM must be filled.

CAUTION: OBSERVE THE LEVEL OF THE TANK CONTENT DURING THIS PROCEDURE!

If necessary top up so that no air is sucked in.

10. Fill the counter pressure side (piping system B) in a similar way and rinse it with 4 times of the filling volume. During this procedure the piston may be left in the retracted position.

Now the complete hydraulic system is filled with oil and the actual bleeding process can be started.

The **bleeding of the hydraulic system** may only be carried out or supervised by trained technicians.

1. Using the pump of the HSLM a pressure of 5 to 6 bar is exerted on the piston side of the spring cylinders, which should make the pistons or piston rods move out. At this pressure, which can be read at the hydraulic cabinet, each cylinder must be bled until bubble-free oil comes out.
Should the pressure fall during bleeding, it must be corrected by switching on the pump until it again reaches the value quoted above.
After the hydraulic cylinder, the piston accumulators must also be bled individually.

Caution: Depending on the project, it is also possible that the hydraulic pipes may be equipped with separate bleeding valves (e.g. miniature measuring connections for additional manual measurements).
These pipes must also be bled under a pressure of a maximum of 5 to 6 bar.

After this bleeding process has been carried out several times on the piston side of the cylinder and on the piston accumulators, and oil free of air bubbles leaves during bleeding, the pressure from the lifting pressure side is to be removed.

2. The bleeding process on the rod side of the spring cylinders must be carried out using the same procedure as described in paragraph 1. During the procedure the piston/piston rod should be retracted again.
3. Note: Care should be taken that during bleeding no unnecessarily high pressures are used.
4. After completion of the bleeding process, the piston should be moved in and out several times. Afterwards the fluid system needs enough time to settle and to separate out any remaining air. Only after this has happened should a further bleeding procedure be carried out.
5. The bleeding procedure described above has to be repeated during the commissioning stage until it is certain that the complete hydraulic system does not contain any air bubbles.
6. Before or during bleeding the following points must be checked:
 - The suction pipe inside the HSLM tank must be constructed in such a way that even in difficult operational situations (at maximum operating pressure and with lifted rollers) no air is sucked in.
 - during the bleeding, the hydraulic system must be checked for tightness (screw connections etc).